Koustav Chandra

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Research Interests

Gravitational-wave astronomy and data analysis — compact binary searches and source characterisation; black hole ringdown modelling and spectroscopy; parameterised and population-level tests of general relativity; and the science case for next-generation ground-based gravitational-wave detectors, including Cosmic Explorer, LIGO-India, and the Einstein Telescope.

Education

Indian Institute of Technology, Bombay — Mumbai, IN
Ph.D. in Physics (Supervisor: Prof Archana Pai) — 2018–2023

- Thesis: "Hearing gravitational waves from intermediate-mass black hole binaries: Detection and Characterisation."

National Institute of Technology, Rourkela — Rourkela, IN
Integrated M.Sc. in Physics and Astronomy — 2013–2018

- Thesis: "An Algebraic Study of SO(10) Grand Unified Theory"

Experience

Max-Planck-Institut für Gravitationsphysik (Albert Einstein Institute), Potsdam — Potsdam, DE
Postdoctoral Scholar, Astrophysical and Cosmological Relativity Division — 2025–Current

Institute for Gravitation & the Cosmos, The Pennsylvania State University — State College, US
Postdoctoral Scholar (Mentor: Prof B. S. Sathyaprakash) — 2023–2025

Mentoring

- Mr Leo Ng (Penn State University) — 2026–Current
  PhD candidate. Co-supervising a project on a modelled binary black hole search.
- Mr Giovanni Benetti (Università di Padova) — 2025–2026
  Master's student. Supervised a project on the impact of systematics on golden dark siren cosmology. Preprint: arXiv: 2602.14898
- Mr Johann Fernandes (IIT Bombay) — 2025–Current
  PhD student. Co-supervising a forecasting study of eccentric black hole binaries with a next-generation ground-based detector network; co-supervised his contribution to the eccentricity analysis of GW231123. Resulted in publication: Mon. Not. Roy. Astron. Soc. 553 (2026) stag1934.
- Ms Akshita Mittal (Gran Sasso Science Institute, Italy) — 2024–Current
  PhD student (initially mentored as an undergraduate). Supervised a project assessing the impact of eccentricity on full and ringdown-only analysis. Resulted in publication: Mon. Not. Roy. Astron. Soc. 553 (2026) stag1934.
- Ms Shiksha Pandey (Penn State University) — 2024
  Undergraduate research assistant. Co-supervised project assessing the role of LIGO-India in multi-messenger astronomy for next-generation detectors. Resulted in publication: Astrophys. J. Lett. 985 (2025) L17.
- Mr Sayantan Ghosh (IIT Bombay) — 2023–2024
  PhD student. Developed coherence tests to distinguish noise transients from intermediate-mass black hole binary signals by assessing posterior consistency across detectors. Resulted in publication: Phys. Rev. D 109, 064015 (2024)
- Ms Kritti Sharma (IIT Bombay) — 2022–2023
  Undergraduate student. Developed a transfer learning framework to detect mass-asymmetric black hole binaries in Advanced LIGO data. Resulted in publication: Phys. Rev. D 108 124061 (2023)
- Mr Sagar Gupta (IIT Bombay) — 2022–2023
  Undergraduate student. Implemented a chirp-mass–based classifier for glitch identification in gravitational-wave data, improving transient search data-quality vetoes.

Talks, Lectures & Seminars

♠ **From a whisper to a continuous roar** October 2026
Department of Physics, Indian Institute of Technology Bombay, Mumbai, India

♮ **Selected topics from the Compact Binary Gravitational Wave landscape** April 2026
IMPRS Lecture Week "Gravity at the Extreme: from Theory to Observation", Wandlitz, Germany

♠ **The coming flood of Gravitational Waves in the Mid 2030s: Myths, Forecasts and Surprises** March 2026
Instituto Galego de Física de Altas Enerxías, Universidade de Santiago de Compostela, Spain

♠ **GW231123: The Shape-Shifting Gollum** August 2025
Presidency University, Kolkata & Jawaharlal Nehru University, Delhi, India

♠ **Too Many, Too Long, Too Loud: CBC Data Analysis Challenges in the XG Era** May 2025
PAX-X and CE Symposium, University of Illinois Urbana-Champaign, USA

• **Exploring the XG Era: Foreground Noise Impact & LIGO India's Role** November 2024
Detection and Analysis of Gravitational Waves in the Era of Multi-Messenger Astronomy, Banff, Canada

♠ **Preparing for a Gravitationally-Bright Tomorrow** August 2024
Department of Physics, Indian Institute of Technology Bombay, Mumbai, India

• **Intermediate-Mass Binary Black Holes and How to Find Them** February 2023
Inference-Data-Experiments-Analysis (IDEA) Talks, TIFR, Mumbai, India

• **Prospects of Observing Mass-Asymmetric IMBH Binaries with A+ Sensitivity** May 2023
Science with LIGO-India Meeting, IUCAA, Pune, India

• **Hunting for IMBHs with the International Gravitational-Wave Observatory Network** February 2022
Chennai Symposium on Gravitation and Cosmology, Chennai, India

○ **Inspiral-Merger Informed Ringdown Analysis of Black Hole Merger Remnants** March 2025
APS Global Physics Summit, Anaheim, USA

○ **Preparing for Tomorrow: Simulating the Challenges of Next-Generation Detectors** April 2024
APS April Meeting, Sacramento, USA

○ **Searching for Gravitational-Wave Higher-Order Modes from Asymmetric IMBH Mergers** July 2022
23rd International Conference on General Relativity and Gravitation (GR23), Liyang, China

♮ Invited Lecture ♠ Invited Seminar • Invited Talk ○ Selected Contributed Talk

Collaboration Membership, Organisation & Service

• **Member, LIGO Scientific Collaboration and Cosmic Explorer Consortium** 2019–Current
Paper-writing team for GW231123 and GW230814; lead PyCBC-IMBH analyst for the O3 IMBH search; contributor to the GW190521 discovery paper (details under Publications).

• **Co-lead, CBC Working Group, Cosmic Explorer Science Traceability Matrix** 2024–Current
Forecasts the observatory's black hole and neutron star science across detector design choices, informing funding-agency decisions on XG detector design; maintain the group's public analysis dataset. Also contribute to the Cosmology Working Group.

• **Lead organiser, Cosmic Explorer Science Traceability Matrix Workshop** May 2025
Institute for Gravitation & the Cosmos, Penn State. Co-organisers: V. Prasad and C. Shaffer.

• **Lead organiser, XG Mock Data Challenge Workshop** June 2024
Institute for Gravitation & the Cosmos, Penn State. Co-organisers: I. Gupta, B. S. Sathyaprakash, S. Vitale and D. Brown.

Research Support

• **NSF PHY-2207638 (PI: B. S. Sathyaprakash, Penn State; $330k)** 2023–2025
A Data Challenge for the Next Generation of Ground-Based Gravitational Wave Detectors. Postdoctoral researcher and sole developer of `gwforge`*, the mock-data framework underpinning the challenge.*

• **NSF PHY-2307147 (PI: B. S. Sathyaprakash, Penn State; $371k)** 2023–2025
Understanding Compact Binary Formation with Gravitational Wave Observations. Postdoctoral researcher; led or mentored three publications under this award (LIGO-India in the XG era, GW230529, foreground effects on inference).

Outreach & Science Communication

• **LIGO Science Summaries** 2022, 2025
Wrote the public-facing summaries of two LVK results: GW231123 and the O1–O2 intermediate-mass black hole search.

• **Bayesics for Transient Gravitational Wave Astronomy** December 2024
PUG Seminar, Penn State. Introduced Bayesian inference to undergraduates, graduate students and postdocs through a binary black hole merger.

• **Gravitational Wave Searches for Compact Binary Mergers** May 2023
Gravitational Wave Open Data Workshop #6 (online). Lecture on how binary mergers are detected in LIGO–Virgo data.

- **Gravitational Waves 101** May 2019
  Vigyan Samagam, Nehru Science Centre, Mumbai. Public talk introducing gravitational-wave astronomy.

Academic Awards & Fellowships

- **Naik and Rastogi Excellence in Ph. D. Thesis Award**, Indian Institute of Technology, Bombay 2024
- **Doctoral Research Fellowship**, Ministry of Human Resource Development, Government of India 2018–2023
- **Department Prize for Outstanding Student Performance**, National Institute of Technology, Rourkela 2016

Technical Skills & Software

- **Languages:** Python, C/C++, Bash, LaTeX
- **Libraries:** `gwforge` (lead developer and maintainer; adopted by the Cosmic Explorer consortium for its Science Traceability Matrix), `trill` (lead developer and maintainer), PyCBC (contributor), TEOBResumS (contributor), PyART (contributor)

Publications

28 short-author papers; the LVK list below is curated. Full list at inspirehep.net/authors/1817603. ⊛ co-authored with PhD supervisor; underlined students I mentored.

Short-Author Publications

[1] G. Benetti, **K. Chandra**, and B. S. Sathyaprakash, "Tarnished by Tools: Cost of Systematics in Golden Dark Siren Cosmology", Feb. 2026. arXiv: 2602.14898 [gr-qc].

[2] J. Calderón Bustillo, A. del Rio, N. Sanchis-Gual, and **K. Chandra**, "Intrinsic Handedness in O1-O4a Black Hole Mergers: Probing Orbital Precession, Remnant Retention in Dense Environments, and Cosmological Mirror Asymmetry", *Astrophys. J. Lett.*, vol. 1008, no. 2, p. L38, 2026. arXiv: 2606.17752 [gr-qc].

[3] **K. Chandra** and J. Calderón Bustillo, "Testing General Relativity with GWTC-4.0 through mixture models", Jul. 2026. arXiv: 2607.07070 [gr-qc].

[4] **K. Chandra**, J. Fernandes, A. Mittal, and G. Carullo, "Eccentricity in Disguise? Insights from GW231123 and Numerically Simulated Binary Black Hole Merger Signals", *Mon. Not. Roy. Astron. Soc.*, vol. 553, no. 1, stag1934, 2026. arXiv: 2606.15150 [gr-qc].

[5] D. Chiaramello, N. Cibrario, J. Lange, **K. Chandra**, R. Gamba, R. Bonino, and A. Nagar, "Theory-agnostic tests of general relativity with a new parametrized model for gravitational waves from eccentric, precessing binary black holes", *Phys. Rev. D*, vol. 114, no. 4, p. 044 004, 2026. arXiv: 2511.19593 [gr-qc].

[6] E. Shukla, A. Rashti, R. Gamba, D. Radice, and **K. Chandra**, "GR-Athena++ simulations of spinning binary black hole mergers", *Class. Quant. Grav.*, vol. 43, no. 6, p. 065 019, 2026. arXiv: 2510.13963 [gr-qc].

[7] J. Calderón Bustillo, S. H. W. Leong, and **K. Chandra**, "A complete measurement of a black-hole recoil through higher-order gravitational-wave modes", *Nature Astron.*, vol. 9, no. 10, pp. 1530–1540, 2025. arXiv: 2211.03465 [gr-qc].

[8] J. Calderón Bustillo, A. del Rio, N. Sanchis-Gual, **K. Chandra**, and S. H. W. Leong, "Testing Mirror Symmetry in the Universe with LIGO-Virgo Black-Hole Mergers", *Phys. Rev. Lett.*, vol. 134, no. 3, p. 031 402, 2025. arXiv: 2402.09861 [gr-qc].

[9] **K. Chandra**, "gwforge: a user-friendly package to generate gravitational-wave mock data", *Class. Quant. Grav.*, vol. 42, no. 2, p. 025 003, 2025. arXiv: 2407.21109 [gr-qc].

[10] **K. Chandra** and J. Calderón Bustillo, "Black-hole ringdown analysis with inspiral-merger informed templates and limitations of classical spectroscopy", Sep. 2025. arXiv: 2509.17315 [gr-qc].

[11] **K. Chandra**, R. Gamba, and D. Chiaramello, "From Source Properties to Strong-Field Tests: a multipronged analysis of GW250114 with an effective one-body model for generic orbits", Dec. 2025. arXiv: 2512.04593 [gr-qc].

[12] J. Fernandes, A. Pai, and **K. Chandra**, "Improving the detection significance of gravitational wave transient searches with convolutional neural network models", *Phys. Rev. D*, vol. 112, no. 8, p. 083 043, 2025, ⊛. arXiv: 2505.08332 [gr-qc].

[13] I. Gupta, **K. Chandra**, and B. S. Sathyaprakash, "Foreground signals minimally affect inference of high-mass binary black holes in next-generation gravitational-wave detectors", *Phys. Rev. D*, vol. 111, no. 10, p. 104 013, 2025. arXiv: 2410.22302 [gr-qc].

[14] S. Khadkikar, I. Gupta, R. Kashyap, **K. Chandra**, R. Gamba, and B. S. Sathyaprakash, "Precise and accurate neutron star radius measurements with next-generation gravitational wave detectors", *Phys. Rev. D*, vol. 112, no. 6, p. 063 020, 2025. arXiv: 2502.03463 [astro-ph.HE].

[15] S. Pandey, I. Gupta, **K. Chandra**, and B. S. Sathyaprakash, "The Critical Role of LIGO-India in the Era of Next-generation Observatories", *Astrophys. J. Lett.*, vol. 985, no. 1, p. L17, 2025. arXiv: 2411.10349 [gr-qc].

[16] A. Rashti, R. Gamba, **K. Chandra**, D. Radice, B. Daszuta, W. Cook, and S. Bernuzzi, "Binary black hole waveforms from high-resolution gr-athena++ simulations", *Phys. Rev. D*, vol. 111, no. 10, p. 104 078, 2025. arXiv: 2411.11989 [gr-qc].

[17] **K. Chandra**, I. Gupta, R. Gamba, R. Kashyap, D. Chattopadhyay, A. Gonzalez, S. Bernuzzi, and B. S. Sathyaprakash, "On the Origins, Remnant, and Multimessenger Prospects of the Compact Binary Merger GW230529", *Astrophys. J.*, vol. 977, no. 2, p. 167, 2024. arXiv: 2405.03841 [astro-ph.HE].

[18] **K. Chandra**, A. Pai, S. H. W. Leong, and J. Calderón Bustillo, "Impact of Bayesian priors on the inferred masses of quasicircular intermediate-mass black hole binaries", *Phys. Rev. D*, vol. 109, no. 10, p. 104 031, 2024, ⊛. arXiv: 2309.01683 [gr-qc].

[19] S. Ghosh, **K. Chandra**, and A. Pai, "Unmasking noise transients masquerading as intermediate-mass black hole binaries", *Phys. Rev. D*, vol. 109, no. 6, p. 064 015, 2024, ⊛. arXiv: 2312.01211 [gr-qc].

[20] J. Calderón Bustillo, N. Sanchis-Gual, S. H. W. Leong, **K. Chandra**, A. Torres-Forne, J. A. Font, C. Herdeiro, E. Radu, I. C. F. Wong, and T. G. F. Li, "Searching for vector boson-star mergers within LIGO-Virgo intermediate-mass black-hole merger candidates", *Phys. Rev. D*, vol. 108, no. 12, p. 123 020, 2023. arXiv: 2206.02551 [gr-qc].

[21] J. Calderón Bustillo, I. C. F. Wong, N. Sanchis-Gual, S. H. W. Leong, A. Torres-Forne, **K. Chandra**, J. A. Font, C. Herdeiro, E. Radu, and T. G. F. Li, "Gravitational-Wave Parameter Inference with the Newman-Penrose Scalar", *Phys. Rev. X*, vol. 13, no. 4, p. 041 048, 2023. arXiv: 2205.15029 [gr-qc].

[22] K. Sharma, **K. Chandra**, and A. Pai, "Searching for massive black hole binaries with a transfer learning algorithm", *Phys. Rev. D*, vol. 108, no. 12, p. 124 061, 2023, ⊛. arXiv: 2208.02545 [astro-ph.HE].

[23] **K. Chandra**, J. Calderón Bustillo, A. Pai, and I. W. Harry, "First gravitational-wave search for intermediate-mass black hole mergers with higher-order harmonics", *Phys. Rev. D*, vol. 106, no. 12, p. 123 003, 2022, ⊛. arXiv: 2207.01654 [gr-qc].

[24] J. Calderón Bustillo, S. H. W. Leong, **K. Chandra**, B. McKernan, and K. E. S. Ford, "GW190521 as a black-hole merger coincident with the ZTF19abanrhr flare", Dec. 2021. arXiv: 2112.12481 [gr-qc].

[25] **K. Chandra**, A. Pai, V. Villa-Ortega, T. Dent, C. McIsaac, I. W. Harry, G. S. C. Davies, and K. Soni, "Salient features of the optimised PyCBC IMBH search", in *16th Marcel Grossmann Meeting on Recent Developments in Theoretical and Experimental General Relativity, Astrophysics and Relativistic Field Theories*, ⊛, Oct. 2021. arXiv: 2110.01879 [gr-qc].

[26] **K. Chandra**, V. Villa-Ortega, T. Dent, C. McIsaac, A. Pai, I. W. Harry, G. S. C. Davies, and K. Soni, "An optimized PyCBC search for gravitational waves from intermediate-mass black hole mergers", *Phys. Rev. D*, vol. 104, p. 042 004, 2021, ⊛. arXiv: 2106.00193 [gr-qc].

[27] N. Bose, A. Pai, **K. Chandra**, and V. Gayathri, "Chirp mass based glitch identification in long-duration gravitational-wave detection", *Phys. Rev. D*, vol. 102, no. 8, p. 084 034, 2020, ⊛. arXiv: 2007.03623 [gr-qc].

[28] **K. Chandra**, V. Gayathri, J. Calderón Bustillo, and A. Pai, "Numerical relativity injection analysis of signals from generically spinning intermediate mass black hole binaries in Advanced LIGO data", *Phys. Rev. D*, vol. 102, no. 4, p. 044 035, 2020, ⊛. arXiv: 2002.10666 [astro-ph.CO].

LVK Collaboration Publications with Significant Contributions

[1] A. G. Abac et al., "GW230814: investigation of a loud gravitational-wave signal observed with a single detector", *Astrophys. J. Lett.*, vol. 1004, no. 2, p. L23, 2026. arXiv: 2509.07348 [gr-qc],
Paper-writing team member: coordinated the search, parameter-estimation and tests-of-GR teams, wrote nearly 20% of the manuscript and made nearly all figures.

[2] A. G. Abac et al., "GW231123: A Binary Black Hole Merger with Total Mass 190–265 $M_\odot$", *Astrophys. J. Lett.*, vol. 993, no. 1, p. L25, 2025. arXiv: 2507.08219 [gr-qc],
Fifth-most-cited paper published in Astrophys. J. Lett. during 2024–2026. Paper-writing team member: wrote over 10% of the manuscript, coordinated the search, population, and ringdown teams, and helped reconcile results across pipelines and interpret the ringdown and waveform-systematics studies.

[3] R. Abbott et al., "Search for intermediate-mass black hole binaries in the third observing run of Advanced LIGO and Advanced Virgo", *Astron. Astrophys.*, vol. 659, A84, 2022. arXiv: 2105.15120 [astro-ph.HE],
Led the PyCBC-IMBH analysis with the pipeline I developed, served as primary analyst for the merger-rate estimates and produced nearly all figures.

[4] R. Abbott et al., "GW190521: A Binary Black Hole Merger with a Total Mass of $150M_\odot$", *Phys. Rev. Lett.*, vol. 125, no. 10, p. 101 102, 2020. arXiv: 2009.01075 [gr-qc],
Second-most-cited paper published in Phys. Rev. Lett. during 2019–2026. Contributed to the PyCBC search analysis of the event.